

Session 1 Course Note: AI and the Changing Industrial Workplace

For students in Innovation and Technology Management

Workshop connection: Session 1, 09:00–09:50

Purpose: Build a shared language for discussing AI as an industrial, organizational, and management issue.

How to use this note

Read this note as an introduction rather than as a catalogue of tools. Your goal is to learn how to move from a broad statement such as “we should use AI” to a more useful discussion about a capability, a task, a user, a workflow, evidence, and consequences.

After reading, you should be able to:

- Distinguish an AI model from a complete AI system.
- Compare predictive, classificatory, optimization, computer-vision, generative, and agentic capabilities.
- Identify possible AI roles across an industrial value chain.
- Explain why AI adoption changes work and organizations, not only software.
- Judge an AI opportunity in relation to both expected value and the consequence of error.

1. AI is a system, not a magic object

The term **artificial intelligence** covers many techniques and applications. The OECD describes an AI system as a machine-based system that infers from inputs how to produce outputs—such as predictions, content, recommendations, or decisions—that can affect physical or digital environments. AI systems can differ in autonomy and in whether they adapt after deployment (OECD, 2024 (<https://oecd.ai/en/ai-publications/explanatory-memorandum-on-the-updated-oecd-definition-of-an-ai-system>)).

Three distinctions are important:

1. **A model** is a mathematical or computational component that produces an inference.
2. **An application** combines one or more models with an interface, data, rules, and software.
3. **An organizational system** also includes users, managers, procedures, suppliers, controls, incentives, and affected people.

A language model can draft a maintenance instruction, but the complete work system determines:

- Which documents the model may use.
- Whether the documents are current.
- Who sees the draft.
- What the human must check.
- Whether the draft can trigger an order or equipment change.
- How errors are reported and corrected.
- Who remains responsible.

This is why a technically impressive model can produce poor organizational results, while a modest model can create substantial value in a carefully designed workflow.

2. A practical map of AI capabilities

You do not need to begin with algorithms. Begin with what the system is expected to do.

Capability	Typical output	Industrial example	Main management question
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Prediction	A probability or expected value	Failure risk, demand, delivery delay	Is the prediction sufficiently reliable and timely to change a decision?
Classification or detection	A category or alert	Defect type, safety event, fraud signal	What are the consequences of false positives and false negatives?
Optimization	A recommended allocation or plan	Production schedule, route, energy use	What objective is optimized, and which constraints or stakeholders are missing?
Computer vision	Information inferred from images or video	Surface inspection, object counting, PPE detection	Does performance remain acceptable under real lighting, angles, equipment, and populations?
Generative AI	New text, code, image, audio, or design	Drafting, summarizing, translating, coding, design exploration	Is the output grounded, permitted, checked, and fit for its purpose?
Agentic AI	A sequence of planned digital actions	Collect information, update systems, contact suppliers	Which actions are authorized, reversible, logged, and subject to approval?

These categories can overlap. A service assistant may retrieve documents, classify a question, generate a response, and recommend an action. The more capabilities are connected, the more important it becomes to understand the complete chain.

Generative AI is only one part of AI

Generative AI receives disproportionate attention because it is accessible through natural language. It is highly relevant to knowledge work, but it is not automatically the best solution for an industrial problem.

For example, if the goal is to assign safety reports to stable categories, a focused classifier may be easier to evaluate and control than a general-purpose chatbot. DNV describes a Green Cargo case in which machine learning was used to improve the classification of safety and quality reports. The reported value came from a defined task, operational data, continuous improvement, and measurable changes—not from using the most general technology available (DNV case study (<https://www.dnv.com/cases/green-cargo-uses-ai-to-improve-data-quality-and-efficiency-245284/>)).

3. Where AI may contribute in an industrial company

An industrial organization combines physical operations, engineering knowledge, commercial work, regulation, and long-term relationships. Possible AI applications therefore extend across the value chain.

Product development and engineering

- Search and compare requirements.
- Explore design alternatives.
- Support simulation and testing.
- Detect inconsistencies in specifications.
- Retrieve lessons from previous projects.

The danger is that a plausible suggestion may violate an unstated constraint. Engineering use therefore requires configuration control, approved sources, domain review, and traceability.

Production, quality, and safety

- Inspect components using images or sensor data.
- Detect anomalies in process signals.
- Classify deviations and non-conformances.
- Identify patterns across incident reports.

- Support—but not invent—root-cause hypotheses.

A detected correlation is not automatically a cause. Management must also consider whether workers will change reporting behaviour when an AI system is introduced.

Maintenance and field service

- Predict when equipment may require attention.
- Prioritize inspections.
- Retrieve procedures and service histories.
- Draft a service summary.
- Recommend possible diagnostic steps.

The cost of a missed failure may be much higher than the cost of an unnecessary inspection. This asymmetry should influence thresholds and review requirements.

Supply chains and projects

- Forecast demand and lead times.
- Monitor supplier and delivery risks.
- Compare contractual requirements.
- Summarize project status.
- Support schedule and inventory decisions.

Optimization can create local improvements while harming the wider system. Reducing inventory may look efficient until variability, geopolitical risk, or service requirements are considered.

Commercial and administrative work

- Draft or compare tender material.
- Translate and adapt communication.
- Retrieve policy or contract information.
- Summarize meetings and documents.
- Support customer-service staff.

This work appears low-risk, but documents may contain confidential information, incorrect references can enter formal decisions, and generated language can hide uncertainty.

Management and people

- Support workforce planning and learning.
- Help employees find internal policies.
- Analyse organizational data.
- Assist recruitment or performance processes.

Applications involving employment, education, access, or evaluation can affect rights and opportunities. They require more demanding evidence, fairness analysis, transparency, and human review than a tool that reformats a meeting agenda.

4. Think in tasks, not entire jobs

AI usually affects a collection of tasks before it replaces an entire occupation. A job may contain:

- Routine information processing.
- Physical work.

- Unstructured problem-solving.
- Relationship-building and negotiation.
- Professional judgment.
- Responsibility for consequences.

Different parts of the same job may be assisted, transformed, or left unchanged. The International Labour Organization's global task-level analysis found that many occupations are exposed to generative AI, but concluded that transformation is more likely than complete replacement for most of them because human input remains necessary (ILO, 2025 (<https://www.ilo.org/publications/generative-ai-and-jobs-2025-update>)).

This does not mean that change will be harmless or equally distributed. Entry-level tasks may change first. Some workers may gain access to expertise, while others may experience increased monitoring, workload, or deskilling. A manager should therefore ask both:

- **What work can the technology perform?**
- **What kind of work and workplace are we creating?**

Recent OECD work emphasizes that most workers will not need advanced AI-development skills. Many will need sound digital and data skills, critical interpretation, problem-solving, managerial competence, and the ability to work responsibly with AI-supported processes (OECD, 2026 (https://www.oecd.org/en/publications/ai-and-skills_f843b352-en.html)). These are central ITM capabilities.

5. Adoption is not the same as value

The 2026 Stanford AI Index reports that organizational AI use continued to grow in 2025, while the use of AI agents remained at an early stage in most business functions. It also reports that productivity gains are strongest in structured, measurable tasks where output can be monitored (Stanford HAI, 2026—Economy (<https://hai.stanford.edu/ai-index/2026-ai-index-report/economy>)).

Treat adoption statistics carefully:

- A person may have access without using the tool.
- Usage may increase without quality improving.
- Time saved in one task may reappear as review or correction work elsewhere.
- A pilot result may not generalize to another task, language, location, or workforce.
- A self-reported productivity estimate is not the same as measured operational output.

Useful evidence should connect the technology to a defined task and outcome. You should distinguish:

1. **Availability:** people have access.
2. **Adoption:** people use it.
3. **Output:** work is produced faster or in greater volume.
4. **Quality:** the work is sufficiently correct and useful.
5. **Outcome:** the workflow, customer, safety, or business result improves.
6. **Impact:** the improvement persists and is attributable to the intervention.

6. Two Norwegian examples and one warning

A bounded assistant: FFI and the Norwegian Navy

FFI, the Norwegian Navy, and a technology partner developed and tested an HR assistant based on retrieval-augmented generation over a purpose-specific collection of HR documents. A pilot preceded a field experiment. Frequent users reportedly answered recurring questions faster, while the study also identified integration, trust, experience, and adoption issues (FFI report 25/008 (<https://www.ffi.no/publikasjoner/arkiv/okt-produktivitet-med-kunstig-intelligens-et-feltekspperiment-med-digital-hr-assistent>)).

The lesson is not that retrieval solves every reliability problem. The useful features are the bounded task, identified users, relevant source collection, staged testing, feedback, and evaluation.

A focused classifier: Green Cargo

In the DNV customer case, Green Cargo applied machine learning to the classification of incident reports. The organization had a visible workflow problem: easier reporting increased volume, but incorrect categorization made investigation harder. The reported solution addressed that bottleneck and used operational measures (DNV (<https://www.dnv.com/cases/green-cargo-uses-ai-to-improve-data-quality-and-efficiency-245284/>)).

This is also a reminder to treat a vendor-published case as evidence from an interested source. It can illustrate implementation but is not independent proof that the same product will create the same result elsewhere.

Plausible but invalid references: Asker municipality

In Asker, AI-assisted legal searching contributed to incorrect or non-existent court references entering a political decision document. Professional-looking language did not make the references valid (VG (<https://www.vg.no/nyheter/i/OkyLKA/asker-kommune-brukte-ai-som-diktet-opp-hoeyesterettsdommer>)).

Calling this only a “hallucination” is incomplete. The organizational questions include the task assigned to AI, access to authoritative legal sources, reviewer competence, time pressure, documentation procedures, and accountability.

7. Consequences determine how much evidence you need

No system is perfectly accurate. Management must decide what kind of error matters and how much evidence is proportionate.

For a binary detection task:

- A **false positive** raises an alert when the condition is absent.
- A **false negative** fails to detect a condition that is present.

The two errors rarely have equal consequences. A false alarm in a low-cost sorting process may be tolerable. Missing a safety-critical defect may not be.

Use four consequence questions:

1. Who benefits if the system works?
2. Who carries the burden if it is wrong?
3. Can the error be detected and reversed?
4. Does the system affect safety, rights, employment, access, or legal obligations?

A useful rule is:

The greater the consequence, autonomy, scale, and difficulty of reversal, the stronger the evidence and controls should be.

8. AI adoption is organizational change

An AI initiative can change:

- The sequence of work.
- Which knowledge is visible.
- Who has authority.
- How performance is measured.
- Which mistakes are likely.
- What employees need to learn.

- How customers or citizens experience the organization.

You should therefore analyse at least four connected layers:

Layer	Example question
Technical	Does the system perform under real operating conditions?
Work process	Where is the output used, checked, corrected, and recorded?
People and organization	Do users have competence, time, authority, and incentives to use it responsibly?
Environment and governance	Which laws, contracts, standards, suppliers, and stakeholders shape acceptable use?

Weak projects often optimize one layer and assume the others will adapt automatically.

9. Your first management checklist

Before supporting an AI proposal, ask:

1. What specific task, decision, or workflow is being changed?
2. Who experiences the present problem?
3. Which AI capability appears relevant, and why is a simpler solution insufficient?
4. What data or documents are required and permitted?
5. What baseline describes current performance?
6. What happens if the system is wrong, unavailable, manipulated, or over-trusted?
7. Which human role checks, decides, overrides, and remains accountable?
8. What small, reversible test could produce credible evidence?

If a proposal cannot answer these questions, it is not yet ready for a technology decision.

10. Reflection questions

1. Choose an industry you know. Identify one task for prediction, one for classification, and one for generative AI. Which creates the greatest consequence if wrong?
2. Where could AI remove routine work but also remove an important learning opportunity for new employees?
3. Find one public AI success claim. Is the evidence about availability, adoption, output, quality, outcome, or impact?
4. In the FFI/Navy case, which features make the assistant more bounded than a public chatbot? Which risks remain?
5. Complete: "AI is likely to create value when ..., but is likely to cause harm when ..."

Key terms

- **AI model:** A computational component that generates an inference from input.
- **AI system:** The model together with data, software, interfaces, rules, people, and operating context.
- **Inference:** The process of generating an output from an input using a trained model or rule system.
- **Generative AI:** AI that produces new content such as text, code, images, audio, or designs.
- **Agentic AI:** AI configured to plan or perform sequences of actions, often using external tools.
- **False positive:** A system reports a condition that is not present.
- **False negative:** A system fails to report a condition that is present.
- **Baseline:** A measurement of present performance used for comparison.
- **Socio-technical system:** A system whose outcomes depend jointly on technology, people,

processes, and institutions.

References and further reading

- OECD. Explanatory memorandum on the updated definition of an AI system (<https://oecd.ai/en/ai-publications/explanatory-memorandum-on-the-updated-oecd-definition-of-an-ai-system>) (2024).
- Stanford Institute for Human-Centered AI. The 2026 AI Index Report (<https://hai.stanford.edu/ai-index/2026-ai-index-report>) and Economy chapter (<https://hai.stanford.edu/ai-index/2026-ai-index-report/economy>) (2026).
- OECD. AI and skills: What we know so far (https://www.oecd.org/en/publications/ai-and-skills_f843b352-en.html) (2026).
- International Labour Organization. Generative AI and jobs: A 2025 update (<https://www.ilo.org/publications/generative-ai-and-jobs-2025-update>) (2025).
- FFI. Økt produktivitet med kunstig intelligens?—et felteksperiment med digital HR-assistent (<https://www.ffi.no/publikasjoner/arkiv/okt-produktivitet-med-kunstig-intelligens-et-felteksperiment-med-digital-hr-assistent>), report 25/008 (2025).
- DNV. Green Cargo uses AI to improve data quality and efficiency (<https://www.dnv.com/cases/green-cargo-uses-ai-to-improve-data-quality-and-efficiency-245284/>).
- USN. Master in Innovation and Technology Management (<https://www.usn.no/english/academics/find-programmes/master-in-innovation-and-technology-management/>).

Session takeaway

Do not begin with “Where can we use AI?” Begin with the work, the people, the evidence, and the consequences. Then decide whether a particular AI capability is appropriate.